



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO R.M.G AGENCY

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

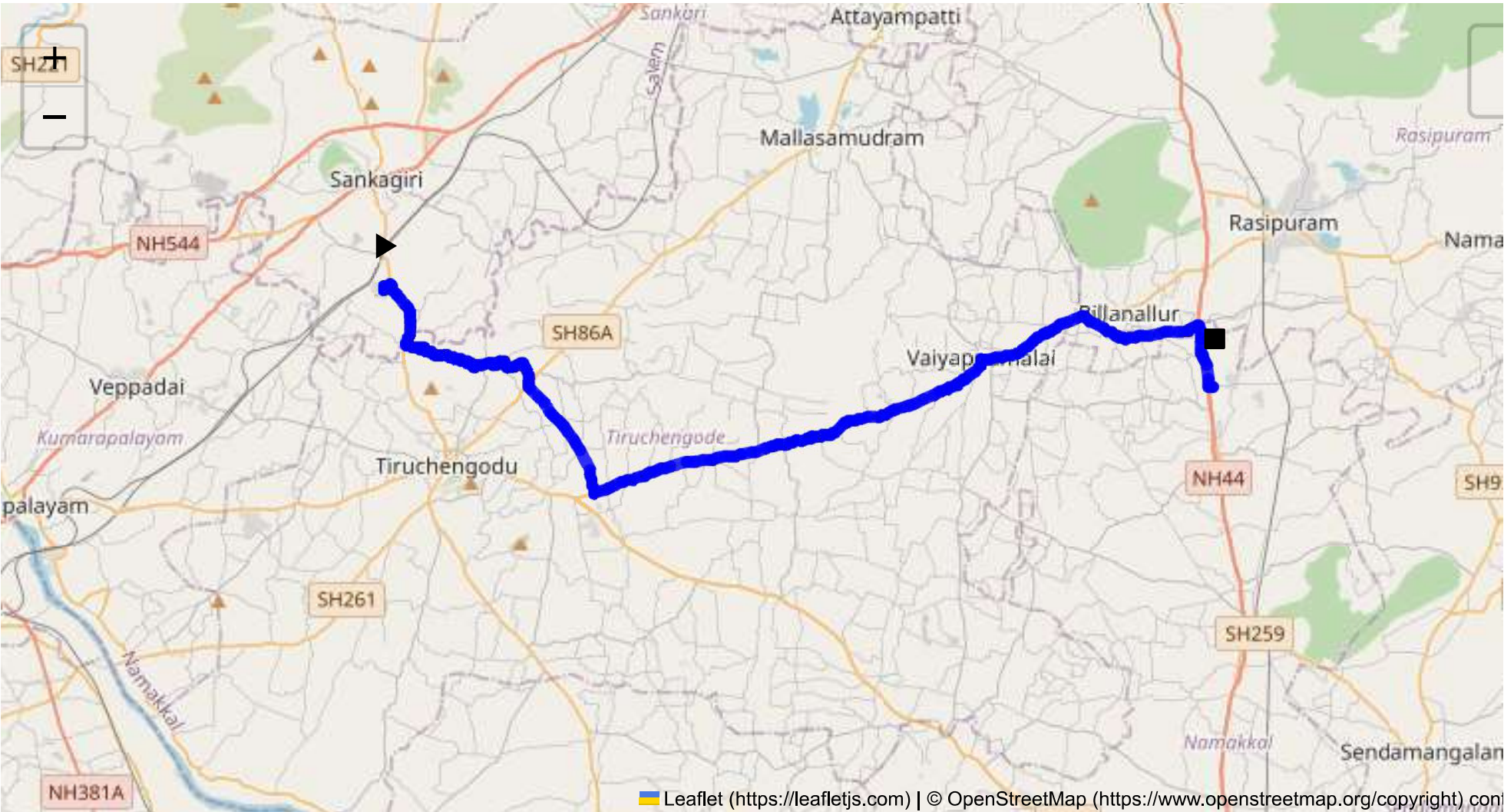
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 41.20 km
Estimated Duration: 0.9 hours
Adjusted Duration (Heavy Vehicle): 1.2 hours
Start: (11.4381, 77.8734)
End: (11.4063657, 78.1610469)



Welcome to the Journey Risk Management Study

Route Safety Analysis Report

Overview of the Route Map: The route from Sangagiri to Ananthakrishnarayasamudram spans approximately 41.20 kilometers. It traverses mainly rural and semi-urban areas, with a mix of highways and local roads.

1. Typical Weather Conditions and Potential Hazards:

- **Climate:** The region experiences a tropical climate. Weather is usually hot and humid, especially in the summer months (April to June).
- **Rainfall:** The monsoon season (June to September) can result in heavy rainfall, causing potential flooding and slippery road conditions.
- **Weather-related Hazards:** Heavy rainfall may lead to waterlogging on poorly drained sections of the road.

2. Analysis of Traffic Patterns:

- **Peak Hours:** Typically, morning (8:00 - 10:00 AM) and evening (5:00 - 7:00 PM) are the busiest due to local commuter traffic.
- **Congestion-prone Areas:** Key intersections close to towns and market areas may experience slow-moving traffic. Sangagiri has known bottlenecks due to local business activities.

3. Assessment of Road Quality and Infrastructure:

- **Highway Sections:** Generally well-maintained but can have potholes due to wear and monsoon damage.
- **Local Roads:** Some sections may have narrow lanes and uneven surfaces. Maintenance varies significantly.
- **Signage and Lighting:** Adequate on main roadways but may be insufficient on some village roads.

4. Suggestions for Alternative Routes:

- **Detours:** In case of emergencies, bypasses exist via SH-86, which may be less congested and provide alternate access to the destination.
- **Road Conditions:** These alternatives should be used with caution during heavy rain due to flooding.

5. Local Regulations Affecting Hazardous Material Transport:

- **Permits:** Ensure possession of all necessary permits for the transport of hazardous materials.
- **Time Restrictions:** Certain areas may have time restrictions for heavy vehicles, particularly during school hours or in high pedestrian zones.

6. Historical Incidents Involving Heavy Vehicles:

- While specific data on incidents is varied, reports indicate occasional accidents in narrow village sections and near construction zones.
- Particular caution is urged near intersections where visibility is limited.

7. Environmental Considerations and Sensitive Areas:

- **Protected Areas:** Proximity to agricultural land means care must be taken in preventing spills.
- **Wildlife Crossings:** Stay alert for animal crossings in rural stretches.

8. Communication Coverage:

- **Network Strength:** Generally good near urban areas but potential dead zones exist in rural sectors.

- **Backup:** Consider using communication devices not reliant entirely on mobile networks.

9. Estimated Emergency Response Times:

- **Urban Areas:** Faster response times typically within 30 minutes.
- **Rural Sections:** Longer response time, potentially up to an hour depending on proximity to emergency services.

10. Overall Summary of Risk Assessment: The route from Sangagiri to Ananthakrishnarayasamudram is generally navigable for heavy vehicles, albeit with some risk areas. Key challenges include traffic congestion during peak hours, potential road damage during monsoon seasons, and limited communication coverage in rural zones. Transports should factor in potential delays and consider alternative routes in case of emergencies. Proper adherence to local traffic regulations and careful planning considering weather forecasts can mitigate most hazards. Overall, with due precautions, the route is manageable for hazardous material transport.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43811, 77.87348	15 KM/Hr
1	Turn	Medium	11.43956, 77.87340	30 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
4	Blind Spot	Blind Spot	11.41975, 77.88075	10 KM/Hr
5	Turn	Medium	11.41876, 77.88487	30 KM/Hr
6	Turn	Medium	11.41829, 77.88568	30 KM/Hr
7	Turn	High	11.41859, 77.88692	15 KM/Hr
8	Turn	Medium	11.41852, 77.88700	30 KM/Hr
9	Turn	Medium	11.41761, 77.88739	30 KM/Hr
10	Turn	Medium	11.41716, 77.88935	30 KM/Hr
11	Turn	Medium	11.41630, 77.89077	30 KM/Hr
12	Turn	Medium	11.41601, 77.89198	30 KM/Hr
13	Turn	Medium	11.41603, 77.89479	30 KM/Hr
14	Turn	Medium	11.41439, 77.90025	30 KM/Hr
15	Turn	Medium	11.41377, 77.90073	30 KM/Hr
16	Turn	Medium	11.41389, 77.90161	30 KM/Hr
17	Turn	Medium	11.41325, 77.90313	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
18	Turn	Medium	11.41276, 77.90349	30 KM/Hr
19	Turn	Medium	11.41227, 77.90470	30 KM/Hr
20	Turn	High	11.41230, 77.90485	15 KM/Hr
21	Turn	Medium	11.41262, 77.90506	30 KM/Hr
22	Turn	Medium	11.41273, 77.90524	30 KM/Hr
23	Turn	High	11.41329, 77.90978	15 KM/Hr
24	Turn	Medium	11.41314, 77.91006	30 KM/Hr
25	Turn	High	11.41313, 77.91057	15 KM/Hr
26	Turn	High	11.41307, 77.91058	15 KM/Hr
27	Turn	Medium	11.41334, 77.91266	30 KM/Hr
28	Turn	High	11.41424, 77.91433	15 KM/Hr
29	Turn	Medium	11.41409, 77.91458	30 KM/Hr
30	Turn	High	11.41208, 77.91551	15 KM/Hr
31	Turn	High	11.41347, 77.92096	15 KM/Hr
32	Turn	Medium	11.41336, 77.92130	30 KM/Hr
33	Turn	High	11.40566, 77.92355	15 KM/Hr
34	Turn	High	11.40557, 77.92343	15 KM/Hr
35	Blind Spot	Blind Spot	11.36920, 77.94568	10 KM/Hr
36	Turn	Medium	11.41332, 78.08072	30 KM/Hr
37	Turn	Medium	11.41463, 78.08067	30 KM/Hr
38	Turn	High	11.42964, 78.11587	15 KM/Hr
39	Blind Spot	Blind Spot	11.42631, 78.15605	10 KM/Hr
40	Turn	High	11.40562, 78.15949	15 KM/Hr

Emergency Locations

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	marketplace	Farmer Market	11.4146849, 78.0805463	30 km/h	Medium
1	school	Government Higher Secondary School	11.4150045, 78.0838172	30 km/h	Medium
2	school	Government school	11.4248027, 78.1323902	30 km/h	Medium
3	college	Paavai Arts And Science College For Women	11.4262443, 78.1546987	30 km/h	Medium
4	college	Gnanamani Institutions	11.4210938, 78.157164	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.41975, 77.88075



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41876, 77.88487



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41829, 77.88568



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.41859, 77.88692



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41852, 77.88700



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41761, 77.88739



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41716, 77.88935



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41630, 77.89077



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41601, 77.89198



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41603, 77.89479



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41439, 77.90025



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41377, 77.90073



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41389, 77.90161



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41325, 77.90313



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41276, 77.90349



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41227, 77.90470



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.41230, 77.90485



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41262, 77.90506



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41273, 77.90524



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.41329, 77.90978



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41314, 77.91006



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.41313, 77.91057



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.41307, 77.91058



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41334, 77.91266



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.41424, 77.91433



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41409, 77.91458



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.41208, 77.91551



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.41347, 77.92096



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41336, 77.92130



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.40566, 77.92355



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.40557, 77.92343



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.36920, 77.94568



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41332, 78.08072



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41463, 78.08067



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.42964, 78.11587



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.42631, 78.15605

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.40562, 78.15949

